

While-Loop

If we **do not know how many iterations will occur**, then we generally use a `while` loop.

A `while` loop runs **until a given condition becomes false**.

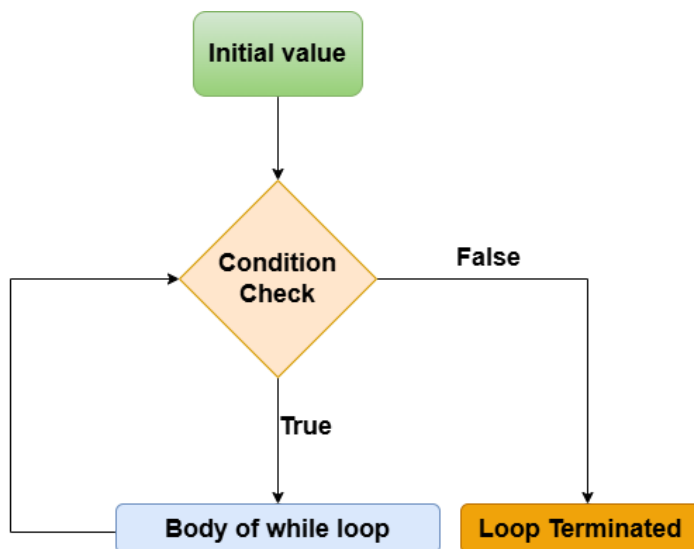
Here, real-life examples of `while` loop in Python** (when number of iterations is unknown).

1.Password Checking System:-

```
password = ""

while password != "admin123":
    password = input("Enter password: ")

print("Access Granted")
```



2. ATM Withdrawal System

```
balance = 1000

while balance > 0:
    amount = int(input("Enter amount to withdraw: "))
    balance = balance - amount
    print("Remaining Balance:", balance)

print("Account Empty")
```

3.Menu Driven Program

```
choice = 0
```

```

while choice != 4:
    print("1. Add")
    print("2. Subtract")
    print("3. Multiply")
    print("4. Exit")

    choice = int(input("Enter choice: "))

print("Program Ended")

```

Examples

Printing numbers from 1 to 5 by using while loop.

```

x=1
while x<=5:
    print(x)
    x+=1

```

Printing numbers from 1 to 5 by using while loop.

```

x=1
while x<=5:
    if x<5:
        print(x,end=",")
    else:
        print(x,end="")
    x+=1

```

Printing even numbers from 10 to 20 by using while loop.

```

x=10
while (x>=10) and (x<=20):
    print(x)
    x+=2

print("End")

```

print sun of given n netural no

```

x=int(input("Enter any no : "))
sum=0
i=1
while i<=x:
    sum=sum+i
    if i<x:
        print(i,end="+")

```

```
else:
    print(i,end=" ")
i=i+1
```

```
print(sum)
```

print n even numbers

```
x= int(input("Enter how many even number you want :"))
n=1
while n<=x:
    print(2*n)
    n=n+1
```

print n even numbers(1,2,3,4,5,6,-----)

```
x= int(input("Enter how many even numbers you want :"))
n=1
while n<=x:
    if n<x:
        print(2*n,end=", ")
    else:
        print(2*n,end=" ")
    n=n+1
```

Print sum of given even numbers.

```
x= int(input("Enter how many even numbers sum you want :"))
n=1
sum=0
while n<=x:
    sum=sum+2*n
    if n<x:
        print(2*n,end="+ ")
    else:
        print(2*n,end=" ")
    n=n+1
print(sum)
```

Print n odd numbers

```
x= int(input("Enter how many odd number you want :"))
n=1
while n<=x:
    if n<x:
        print((2*n-1),end=", ")
    else:
```

```
        print((2*n-1),end="")
n=n+1
```

Print sum of n odd numbers

```
x= int(input("Enter how many odd number you want :"))
n=1
sum = 0
while n<=x:
    sum=sum+(2*n-1)
    if n<x:
        print((2*n-1),end="+")
    else:
        print((2*n-1),end="=")
    n=n+1
print(sum)
```

WAP to print even no upto n netural no.

```
n=int(input("Enter no : "))
i=2
while i<=n:
    if i<=n-2:
        print(i,end=",")
    else:
        print(i)
    i+=2
```

WAP to print odd no upto n netural no.

```
n=int(input("Enter no : "))
i=1
while i<=n:
    if i<=n-2:
        print(i,end=",")
    else:
        print(i)
    i+=2
```